

Mohamed Ashfaq

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Chennai, India | Open to Relocation

SUMMARY

Software Engineer specializing in full-stack development and application security, with hands-on experience building production web and mobile applications, secure REST APIs, and multi-tenant systems. Experienced in React, TypeScript, Python, Node.js, Flutter, PostgreSQL, and Supabase, with expertise in RBAC, JWT authentication, OWASP security practices, and secure API design. RHCSA-certified Linux administrator with experience across software development, cybersecurity, and production system deployment.

SKILLS

Languages: Java, Python, JavaScript, TypeScript, SQL, Dart, C++

Frameworks: React, Flutter, Node.js, Express.js, FastAPI, Tailwind CSS, Capacitor

Tools: Git, GitHub, Postman, Docker, Linux (RHEL, Ubuntu), Nginx, CI/CD

Security: OWASP Top 10, JWT, REST API Security, RBAC, Burp Suite, Wireshark, Nmap, Nessus, Metasploit

Core: Data Structures and Algorithms, OOP, DBMS, Computer Networks, Operating Systems, Agile/Scrum

EXPERIENCE

Software Development Engineer Intern

Dec 2025 – May 2026

Daasoft IT Solutions | Pallavaram, India (Hybrid) | Verification

- Delivered 3+ web and mobile applications using Flutter, PHP, and MySQL, reducing frontend load times by 30% through UI optimization.
- Developed cross-platform Flutter applications and responsive PHP/MySQL dashboards supporting production workflows and 500+ daily API requests.
- Secured 10+ REST API endpoints with JWT-based authentication and authorization controls, addressing BOLA vulnerabilities in application workflows.
- Designed MySQL RBAC schemas for 4+ multi-tenant modules, enforcing role-based permissions and organizational data isolation.
- Worked within a 5-member Agile/Scrum team, using Git for development and Postman and Burp Suite for API testing and security validation.

Tech Stack: PHP, MySQL, Flutter, Dart, JavaScript, REST APIs, JWT, Linux, Git, Postman, Burp Suite

Software Development Engineer Intern

Jun 2026 – Aug 2026

Cognivo-Future Technologies | Chennai, India (Hybrid) | Verification

- Developed full-stack features with React, TypeScript, Node.js, and Express.js across frontend interfaces, backend services, and REST APIs.
- Created Python and FastAPI services connected to Supabase and PostgreSQL for structured data processing and application workflows.
- Designed React Flow interfaces for configurable workflow creation and visual process management within the platform.
- Integrated Razorpay payment workflows and added authentication, authorization, and secure API controls across application services.
- Analyzed competing platforms and translated workflow gaps into product features while debugging, testing, and refining application functionality with the engineering team.

Tech Stack: React, TypeScript, Node.js, Express.js, Python, FastAPI, Supabase, PostgreSQL, React Flow, Razorpay, Capacitor

PROJECTS

AwardX – Global Award Management Platform | GitHub

Tech Stack: React, TypeScript, Tailwind CSS, Node.js, Supabase, PostgreSQL, REST APIs

- Founded and launched AwardX, a multi-tenant platform covering program creation, public submissions, evaluation, certification, analytics, and administration.
- Analyzed global award management platforms to identify workflow gaps and user requirements, shaping product architecture, feature priorities, and roadmap decisions.
- Created a configurable form and workflow engine for defining award categories, eligibility criteria, submissions, evaluation rounds, schedules, reviewer workflows, and certificates without code changes.
- Secured tenant data with RBAC and PostgreSQL Row Level Security (RLS), restricting access across users, organizations, programs, submissions, reviews, and administrative operations.
- Deployed the platform for market validation and continued product development, with current efforts focused on customer acquisition and investment-led market expansion.

ACMH-IDS – UAV CAN Intrusion Detection System | GitHub

- Tech Stack:** Python, PyTorch, Scikit-learn, XGBoost, Optuna, Isolation Forest, Autoencoder, LSTM
- Developed a hybrid IDS combining Isolation Forest, Autoencoder, and LSTM models with XGBoost stacking to detect DoS, Fuzzing, Replay, and Masquerade attacks in UAV CAN networks.
 - Extracted temporal and payload features from time deltas, rolling statistics, payload change rates, transmission frequency, and CAN-ID patterns for anomaly detection.
 - Optimized the fusion pipeline with Optuna and an MCC-optimal decision threshold of 0.794, achieving 93.02% balanced accuracy, 93.09% F1-score, and 99.84% precision.
 - Evaluated the final model using 5-fold time-series cross-validation, achieving 87.20% recall, 98.83% specificity, 0.938 ROC-AUC, and 0.993 PR-AUC.

Rental Utility POS and Inventory Engine | GitHub

- Tech Stack:** React, TypeScript, Tailwind CSS, PHP, MySQL, REST APIs
- Built an asset tracking and point-of-sale system managing 1000+ ledger transactions, delivery challans, and real-time inventory with React and TypeScript.
 - Created dashboards for operational metrics, transport costs, and financial reconciliation to support business reporting and decision-making.
 - Protected backend APIs against XSS and SQL injection using output encoding, input sanitization, parameterized queries, and CSP headers.
 - Deployed the application on Linux/Nginx infrastructure with SSL/TLS certificates, firewall rules, and hardened file permissions.

EduTenant LMS – Enterprise Multi-Tenant Learning Management System | GitHub

- Tech Stack:** Flutter, Dart, PHP, MySQL, Secure REST APIs, State Management
- Built a multi-tenant LMS supporting enrollment, attendance tracking, and grading across 3+ organizational tenants with strict data isolation.
 - Applied database-level tenant isolation and secured REST APIs to prevent unauthorized cross-tenant data access.
 - Optimized MySQL indexes and query execution plans, reducing API response times by 45% for academic progress dashboards.
 - Automated mobile and backend releases through CI/CD pipelines, reducing release cycles by 50% and enabling continuous delivery.

EDUCATION

B.Tech in Computer Science and Engineering – Cyber Security	2023 – 2027
SRM Institute of Science and Technology, Kattankulathur, India	

CERTIFICATIONS

Red Hat Certified System Administrator (RHCSA)	2024
Red Hat – Credential verified via Red Hat Certification Central Verification Link	

C++ Systems Programming Essentials	2024
Scaler Academy Verification Link	